Amphenol Technology (Zhuhai) Co., LTD　　　G8D Report

Customer Part Number	228-1010-9001	**CAR#:** 8D-201910211
Amphenol P/N	C91-147720-16P /(PT07C-20-16P)	**Date：** 20191021 *2020-0318 update*

Complaint Subject

Leak failure& Axial load test failure

Note: reference for 8D-2010905081 into 8D-201910211

D-1 Team Member:

Champion: Carl Chen – Sr. Quality Manager - ATZ

Team Leader: Cary Deng – Customer Quality Engineer

Team Members:

Kam Chen—Assy. Manager

Manqun Tang—Assy. Line Leader

June Zeng---Product Manager

Chris France---Product Manager

Lake Yong---Quality Engineer

Hector Ochoa—QA Manager, Nogales

Xiaolin Wu----IPQC Leader

Briseida Zarate ----Customer Service

DaYu Zhong---ME

Jackson Liu---ME manager

YouChao Liang---molding manager

Beck Gong—Molding ME

D-2 Details Of Complaint

Sep6th, 2019 Celestica directly advised Amphenol about a connector with date code 1916 that failed leak test at Celestica.

D/C:1916

Qty:1

Sep 17th, 2019 Liberty returned connector for leakage test, there are 5 connectors failure

Occurrence site: Axial load test and leakage test per in Amphenol Nogales.

D/C:1916 and 1932

Qty:5

D-3. Containment Action	When	Who	Status
1-Quarantined connectors in Amphenol inventory 0 pcs C91-147720-16P in China, 0pcs in MX.	2019-9-17	Xiaolin	Completed
2- 0pcs products in transit from all Amphenol facilities to Liberty.	2019-9-17	Manqun	Completed
3- Returned 1pcs defect returned to Amphenol in Zhuhai for investigation; Received at Amphenol Zhuhai 10-11-19	2019-10-11	Cary	Completed
4- RMA to return 1pc of 1916 date code with leak failure.	2019-10-11	Hector/Carl	Completed
5-RMA to return all 389pcs in Liberty for screening (in pressure 30psi) in AIO Nogales. 384pcs passed and 5pcs failure as below. D/C1916 – 4/62 failure D/C1928 – 0/201 failure D/C1932 – 1/126 failure	2019-10-11	Hector/Carl	Completed
6-5pcs failure. The defects were supplied to Amphenol Zhuhai for Failure Analysis.	2019-10-11	Hector/Carl	Completed

D-4 Root Cause Analysis:

1. 1916 Failure – 1pc from 9-6-19

A - Rejected sample reviewed:

We got 1pc reject from Liberty for investigation, DC 1916, under PO# 167224. After the connector was re-tested in ATZ, the leakage (little air bubble) come from between insert and shell as below showing.

Note: This connector was before improvement action (D-6 CA 201905081) and 100% inspection in Amphenol Zhuhai with 96kpa (or 14psi).

B - Disassembly the defect:

The connector was disassembled as below; there is insufficient primer around leakage area. This is the same as what we analyzed in May as above.

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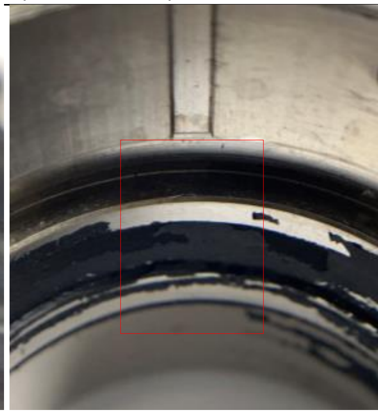


C - Why it was escaped from sorting:

The maximum capacity of test equipment is 100Kpa with MPE +/-2% and test under 96Kpa along with not testing to 9-3353-1 may have contributed to the escape on the failure.

2. 4pcs 1916 Failure & 1pc 1932 Failure

A - Defects come from two date codes as noted as below :

Issue detail	After sorted from Amphenol Nogales	
	Axial load fails (contacts pushed out)	
Date code	1916	1932
QTY	4pcs	1pc

Amphenol Zhuhai performed a leak test on these 5pcs connectors under 30psi and detected no leakage. Therefore, a separate issue has been defined where the contacts have failed Axial Load Testing, not Leak Testing.

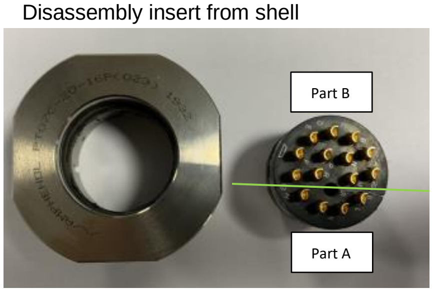
B - For the 5pcs connectors with DC 1916 & 1932, Axial load test failed. This is a separate issue from the 8D 201905081.

B-1 Reviewed the sample, the contact was pushed out as below after Axial Load Test.

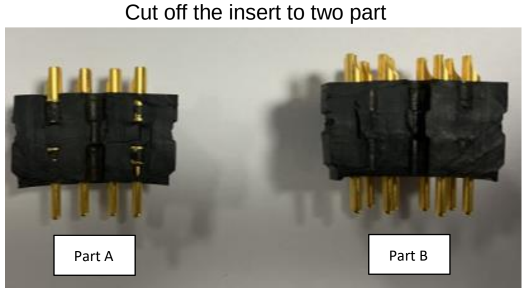
B-2 Disassembled the connector and viewed the glue on the insert and contact.

B-2.1 Disassembled the connector;

Disassembly insert from shell



Cut off the insert to two part



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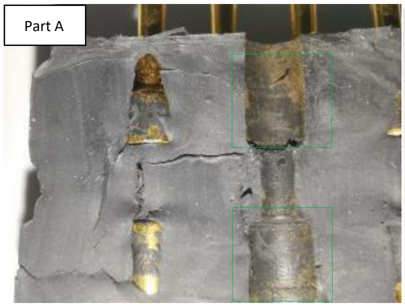
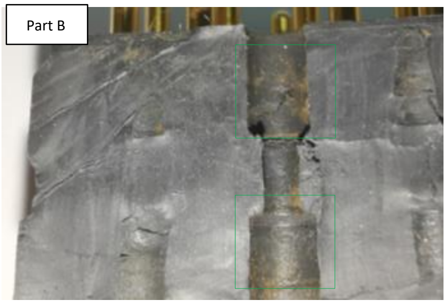
B-2.2 Reviewed on insert, there are glue on the insert;





B-2.3 Reviewed on Contact, there are glue on it as below;

Result: There are enough glue on insert and contacts; this is not believed to be the cause for contact pushed out from insert.

B-3 - Measured the ID of the insert and the OD of the contact.

The ID of the hole on the insert was at the high limit which the spec defined 1.55mm (.061) to 1.65mm (.063), actually it is 1.63mm to 1.65mm, there is a risk of lower retention force. The OD of the contact is in normal level, which spec define 0.066 inch to 0.070 inch, actually 0.068 inches to 0.069.

The target conditions of contact and contact cavity are .066(min) on contact and .065(max) on insert are acceptable.

Note: The insert is molded at Amphenol Zhuhai.

B-4 - Cause analysis:

	1-Why→	**←2-Why→**	**←3-Why→**	**←4-Why→**	**←5-Why**
Direct Cause	Contact was pushed back in the contact cavity from insert;	The bond between insert & shell did not keep the contact stationary for the duration of test	The contact was able to move in the contact cavity	The ID of the contact cavity is at the high limit of the spec.	
Detectability Cause	Amphenol Zhuhai could not detect the failure while Amphenol Nogales reported failure;	Amphenol Zhuhai was sampling inspection in process and Nogales for sorting these returns was 100% inspected by axial load test;	Amphenol Nogales was 100% inspecting by axial load test to follow 9-3353-1; Amphenol Zhuhai was not following 9-3353-1.		
Systemic Clause	Amphenol Zhuhai was not performing the testing per 9-3353-1 in the past	Amphenol Zhuhai did not have 9-3353-1 in our process documentation.	Amphenol Zhuhai had not been shared this process specification	There is no record of this specification being shared with Amphenol Zhuhai	

D-5C/A On Root Cause:

Failure 1: 1- To increase test pressure from 14psi to 30psi for 100% Air Leakage Test plus AQL 1.0 on Cold Temp Soak Test on next 3 lots

Failure 2: 1- To modify mold to guarantee the ID of the hole on the insert at the middle limit;

2- To test 100% of the connector assembly builds on the next 3 lots to 9-3353-1, Test Method B and attach COC data.

3- To full follow spec 9-3353-1 in Amphenol Zhuhai;

4- Amphenol Nogales to test 100% for the next 3 lots assembled at Amphenol Zhuhai to 9-3353-1, Test Method A.

5- Amphenol Nogales to scale down the testing in Action #4 above to AQL stated in 8D- 2010905081 after 3 lots with no failures, and destructive sampling on each lot has been conducted, as well as all reporting of the 100% inspection with no failures is shared with Liberty & Celestica.

D-6 Implemented Permanent Corrective Action：	**When:**	**Who:**	**Follow up Result:**	**By:**
Failure 1 1- Update WI to increase test pressure to 30PSI in Air Leakage Test and follow 9-3353-1 going forward.	10-10	DaYu	Completed	Cary D.
Failure 2 1- To modify the insert mold to the middle limit of the spec;	10-31	YouChao	Completed	
2- To added 100% of the connector assembly builds on the next 3 lots to 9-3353-1, Test Method B and COC will be attached ;	10-23	Manqun/Xiaolin	Completed	
3- To full follow spec 9-3353-1 in Amphenol Zhuhai;	10-10	DaYu	Completed	

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D-7.1 Preventive Action(s) For Similar Process Root Cause Occurrences:	**When:**	**Who:**	**Follow up Result:**	**By:**
Failure 1 – Review Primer Application Process on all PT products；	10-31	DaYu	Completed	Cary D.
Failure 2 – To conduct FA (First Article) process for insert after the mold is to be modified as noted in D-5 and D-6.	11-7	Beck	Completed	

D-7.2 Preventive Action(s) For Similar Process Escape Point Root Cause(s)	**When:**	**Who:**	**Follow up Result:**	**By:**
Failure 1 – Review Leak Test conditions on all PT products;	10-31	DaYu	Completed	Cary D.
Failure 2 - To update QCP, to update the ID of the insert controlling from in spec limit control to in control limit;	10-31	Lake	Completed	

D-7.3 Systemic Prevent Recommendations:	**When:**	**Who:**	**Follow up Result:**	**By:**
Failure 1: 1-Change WI to make sure enough primer application on all PT type C connector assemblies.	10/28	Jackson L.	Completed	Cary D.
		Jackson L.		
2-Update Control Plan & PFMEA t for all PT type C connector assemblies.	10/28	Lake Y.	Completed	
Failure 2: 1-To start FA process for the dimension of the insert after mold was modify;	11-7	Beck	Completed	

Prepared By: Cary Deng
Date: 2019-10-21 2019-10-22update

Approved By(Team leader.): Carl Chen
Date: 2019-10-21 2019-10-22update

LOTS	*LOT #*	*QTY*	*Result*	*Remark*

D-8 Team And Individual Recognition:

［ ］Effective, Close ［ ］Invalidation, See Attachment.

Confirmed By: Approved By: